

LESSON 2-3 • SESSION 2 • PRACTICE • FORM A

Comparing Distributions

Learning target – I can compare two distributions using the median and the interquartile range together. · 6.SP.B.5d

REMEMBER

- **Median** = center: which group is typically higher or lower?
- **IQR** = spread: which group is more **consistent** (smaller IQR)?
- Use **both** to decide. Ask what is better here: a shorter wait or a higher score?

WORKED EXAMPLE

Line 1: median 8 min, IQR 7. Line 2: median 13 min, IQR 12.

Step 1 Center: $13 - 8 = 5$, so Line 1 is typically 5 min shorter.

Step 2 Spread: $7 < 12$, so Line 1 is steadier.

Answer Join Line 1: shorter and steadier

WORDS center (*centro*) · spread (*dispersión*) · distribution (*distribución*)

PRACTICE

show your work · label units

1. **Two Lunch Lines.** Wait times (min). Line 1: 3, 5, 7, 8, 10, 12, 15. Line 2: 2, 4, 11, 13, 14, 16, 20.

Hint Lower median = shorter wait. Smaller IQR = steadier.

	Median	Q1	Q3	IQR
Line 1				
Line 2				

Which to join _____

2. **Two Carrot Plots.** Plot A lengths (cm): 10, 12, 14, 15, 16, 17, 19. Plot B: 5, 11, 14, 15, 20, 23, 27.

a. Median A: ____ b. IQR A: ____ c. Median B: ____ d. IQR B: ____

3. **Two Months.** Month 1: median 120 sales, IQR 40. Month 2: median 135, IQR 12.

a. Typically higher: ____

b. Steadier: ____

4. **Nikhil's Claim.** Set E: median 60, IQR 30. Set F: median 55, IQR 8. Nikhil says Set E wins in every way. Is Nikhil right?

Hint A higher median says nothing about spread.

☐ Right ☐ Not right

5. **Two Water Coolers.** The box plots show cups served per hour.

Hint Read the box edges for Q1 and Q3.

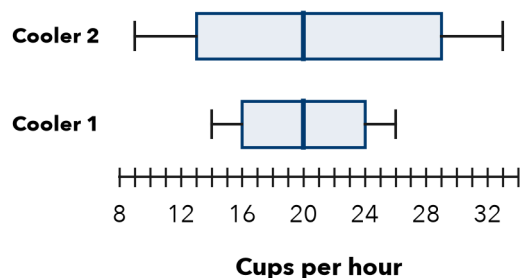
a. Median of 1: _____

b. IQR of 1: _____

c. Median of 2: _____

d. IQR of 2: _____

e. Compare them:



6. **Coach's Choice.** Team 1: median 58 points, IQR 20. Team 2: median 52, IQR 5. The coach wants the team most likely to score close to its usual total.
- ☐ Team 1 ☐ Team 2

Why_____

7. **Two Stores.** Store P: median 46 sales, IQR 9. Store Q: median 52, IQR 21. Write a comparison that uses both measures.

8. **Two Study Methods.** Method 1 quiz scores: median 78, IQR 6. Method 2: median 83, IQR 24.
- a. Higher typical score: _____
- b. More dependable (use both IQRs): _____
- c. Pick one method for the whole class. Argue using both measures.

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Kofi's Work.** Kofi wrote: "Group 1 has the larger median, so Group 1 must have the larger IQR too." Explain the mistake.

TEST PRACTICE

choose the best answer

10. To compare two skewed distributions fully, report the –
- A** median and IQR **B** median only **C** greatest values **D** number of values
11. Set A: median 60, IQR 5. Set B: median 60, IQR 18. Set A is –
- A** higher **B** more consistent **C** more spread out **D** lower

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

Month 2 T	12 N	5 E	Month 1 Z	16 R	Method 2 S	Line 2 Q	8 E	22 X	Line 1 C
#1		#2b	#2d	#3a	#5b	#5d	#8a		